RO2 – Aufgabe: Realisieren Sie einen Prozessor unter Verwendung der Hardware-Modellierungssprache VHDL für den folgenden Befehlssatz:

- Mnemonic: **label: instruction operand1, operand2 ; comment**
- 18-bit instruction width
- register names are represented by ‘sx’ or ‘sy’
- x : Register s0 to sF
- y : Register s0 to sF
- Constant values are represented by ‘aaa’, ‘kk’, ‘ss’, ‘p’ and ‘pp’
- aaa : 12-bit address 000 to FFF
- kk : 8-bit constant 00 to FF
- pp : 8-bit port ID 00 to FF
- p : 4-bit port ID 0 to F
- ss : 8-bit scratch pad location 00 to FF

ADD sX,kk	OUTPUT sX,(sY)
ADD sX,sY	OUTPUT sX,pp
ADDCY sX,kk	RL sX
ADDCY sX,sY	RR sX
AND sX,kk	SL0 sX
AND sX,sY	SL1 sX
COMPARE sX,kk	SLA sX
COMPARE sX,sY	SLX sX
INPUT sX,(sY)	SR0 sX
INPUT sX,pp	SR1 sX
JUMP aaa	SRA sX
JUMP C a a a	SRX sX 0
JUMP NC aaa	SUB sX,kk
JUMP NZ aaa	SUB sX,sY
JUMP Z aaa	SUBCY sX,kk
LOAD sX,kk	SUBCY sX,sY
LOAD sX,sY	TEST sX,kk
OR sX,kk	TEST sX,sY
OR sX,sY	XOR sX,kk
	XOR sX,sY